

Digital Twin for Intersection Throughput and Air Pollution in Hanoi

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Meet the Team



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Mechanical
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Disaster Risk

Overview



**Analyzing Traffic
at Chua Boc
Intersection**



**Three Chosen
Scenarios**



**Scientific
Question**



**Digital Twin
Results**



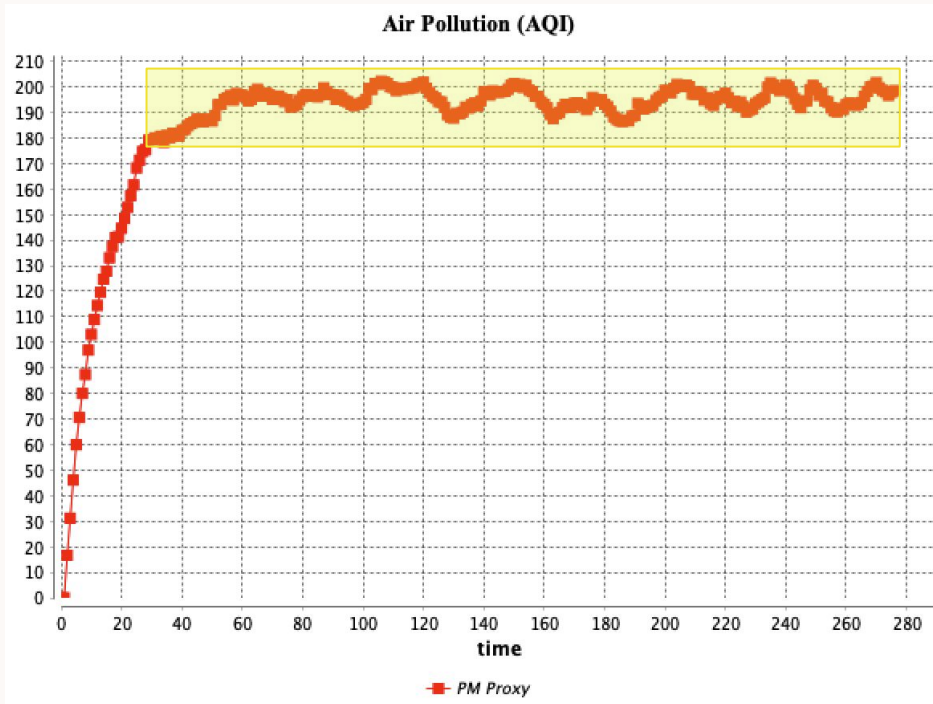
**Digital Twin
Model**



**Summary of
Findings**



Analyzing Traffic at Chùa Bộc Street Intersection



Simulated Air Pollution AQI index over time

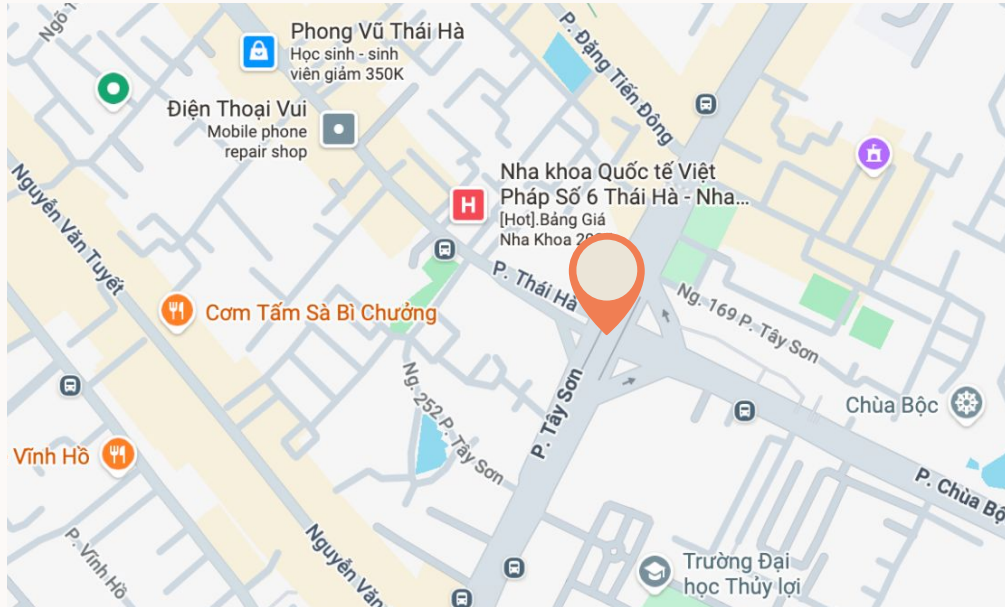
Observed Traffic Dynamics

- Motorbikes dominate traffic and move flexibly across available space, not fixed lanes
- Mixed vehicle types compete for the same road space
- Buses face repeated interruptions, reducing schedule reliability
- Pedestrian crossings add stop and go waves, resulting in inconsistent flow



Usual Daily Traffic

Analyzing Traffic at Chùa Bộc Street Intersection

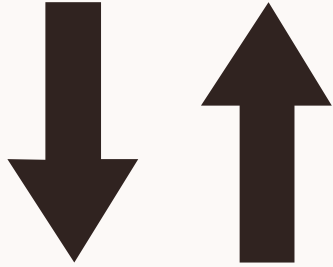


Chùa Bộc Intersection

Why This Intersection Matters

- Represents a high-density, mixed-traffic environment common in central Hanoi
- Congestion is driven by interaction intensity, not signal timing alone
- Small behavioral changes can produce large impacts on throughput and emissions

Scientific Questions



How would the introduction of dedicated bus lanes impact motorcycle usage and public transportation efficiency?



Which vehicle types contribute most significantly to air pollution, and how can this hierarchy inform targeted, step-by-step policy interventions?

Digital Twin Model

AGENTS

Vehicle
-speed -lanes -emission rate -target path



Car: inherits

Motorbike: inherits

Bus: EV & stops

Pedestrian: walk

ENVIRONMENT

Road
-speed allowance -bus only -vehicle separation

Traffic Lights/ Crosswalk
-green or red -cycle time

Bus Stops
- location - traffic dwell

Traffic Police
-check compliance

COMMUNICATION

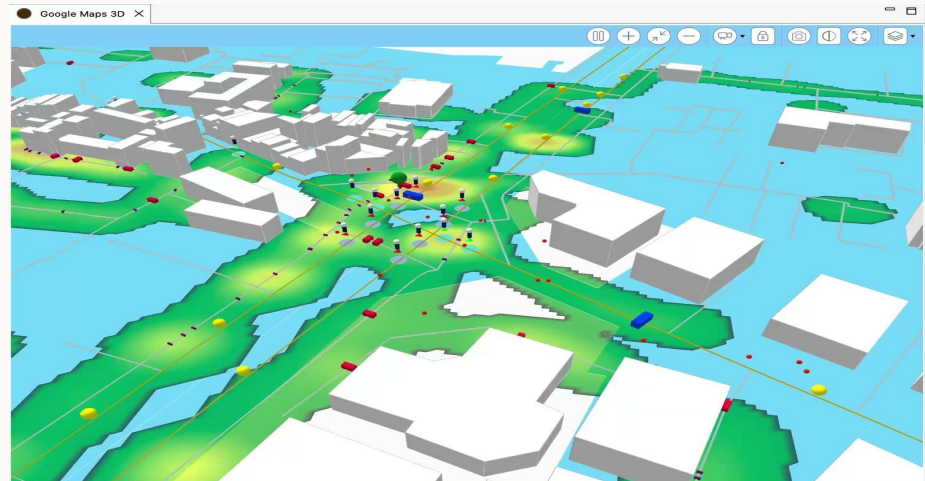
Vehicle → moves on Road

Vehicle → stops at Traffic Light

Bus → moves to + stops at Bus Stops

Vehicle → emits to AQI metric

The vehicles within the environment deliver a **total AQI metric** to represent air quality changes in relation to the baseline case



Three Chosen Scenarios

All scenarios compared to baseline model where lanes do not exist

BUS LANE

Creation of an outer road defined lane for quicker EV public transit



THREE LANES

Having separate defined lanes for all three vehicle types to reduce congestion



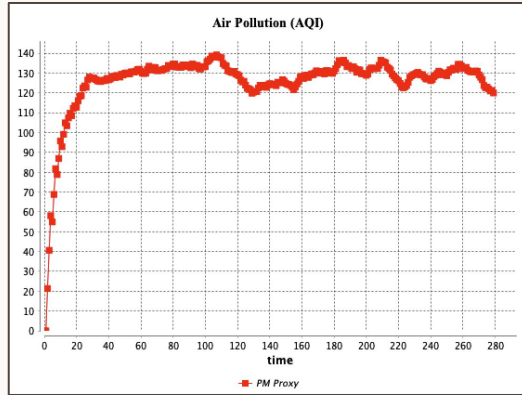
LOW COMPLIANCE

Bus lanes exist with motorbikes still entering & heavy heterogeneity



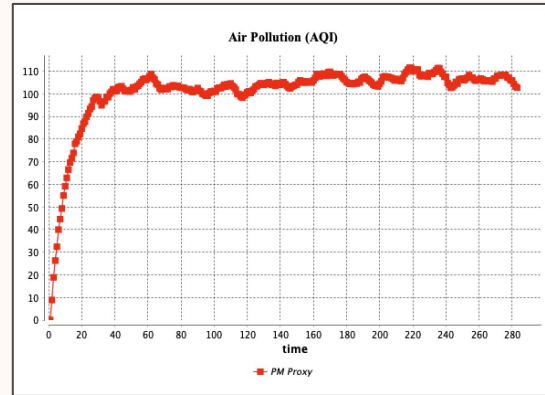
Digital Twin Results

Bus Lane Only



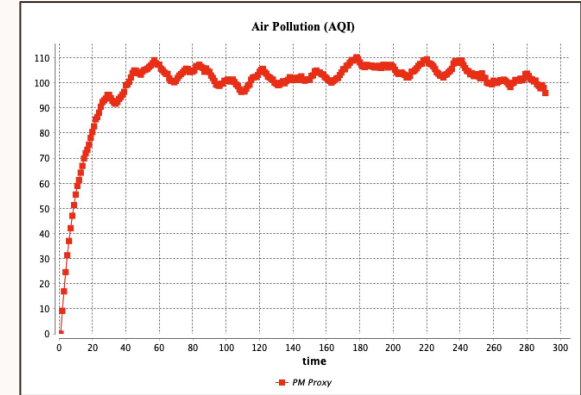
AQI around 130
Third best option

Three Separate Lanes



AQI around 100
Best option

Low Compliance



AQI around 105
Second best option

Low compliance with three separate lanes performs better than only bus lanes & the baseline scenario, demonstrating the importance of policy in optimal air quality improvement.

Summary of Findings

Key Findings

- In our model, bus only lanes increased congestion and worsened air quality, opposite of our hypothesis
- The main pollution driver was stop and go traffic from mixed vehicle interactions
- Traffic flow changes reduced emissions without relying only on electrification or vehicle bans

Policy Implications

- Bus only lanes are low cost, but AQI gains were small in our model
- Separating lanes improves flow even when compliance is imperfect
- Enforcement is needed to sustain benefits over time
- Pilot by corridor first to improve air quality while maintaining access

Suggestion: Improving air quality at congested intersections is best achieved through a sequence of targeted, evidence based interventions, guided by digital twin simulation rather than a single sweeping fix.

THANK YOU!

